Question

A common compound $\mathbf{A}$ is composed of four elements and has a boiling point of $189^{\circ}\mathrm{C}$, with one element accounting for 41.04% of its composition. Compound $\mathbf{B}$ has an empirical formula molecular weight of 63.46. At $195\ \mathrm{K}$, $\mathbf{A}$ and $\mathbf{B}$ are mixed and added to ethanol, followed by the addition of a weak base $\mathbf{C}$ after a while, resulting in the formation of a pungent-smelling liquid $\mathbf{D}$ and a volatile, foul-smelling liquid $\mathbf{E}$.

It is known that the mass fraction of a certain element in $\mathbf{C}$ is 13.86%.

Which of the following statements is correct:

A. All other options are incorrect

B. The chemical equation for the question involves, when balanced with the simplest integer ratio of coefficients, a sum of coefficients for all species less than 10.

C. The chemical equation for the reaction involved in the problem, when balanced with the simplest integer ratio of coefficients, has an even sum of coefficients for all species.

D. The chemical equation for the reaction involved in the problem, when balanced with the smallest possible integer ratio for the coefficients, has a sum of all species' coefficients that is a prime number.

E. The reaction chemical equations involved in the question, after balancing with the simplest integer ratio for coefficients, have a sum of coefficients for all species that is a multiple of 3.

F. The reaction chemical equation involved in the question, after balancing with the simplest integer ratio of coefficients, has a sum of coefficients for all species equal to 10.

G. The reaction chemical equation involved in the problem, after being balanced with the simplest integer ratio as the coefficient sum, has a coefficient sum of 13 for all species.

H. The problem involves the reaction chemical equation, and the products have 4 kinds of gases at $300\ \mathrm{K}$.

I. Cannot use $(COCF_3)_2O$ to replace $\mathbf{B}$ to undergo a similar reaction.

J. Cannot use DCC (dicyclohexylcarbodiimide) to replace $\mathbf{B}$ to undergo a similar reaction to obtain $\mathbf{D}$.

Answer

Correct Answer: D

Detailed Explanation

$\mathbf{B}$'s simplest formula has a molecular weight of 63.46 and can only contain Cl with a atomic weight of 35.45.

CHECKPOINT 1 PTS

$\mathbf{B}$ can only contain Cl with a molecular weight of 35.45

The remaining molecular weight is 28.01, which is obviously CO; (N_2 is 28.02, which does not match). Therefore, the simplest formula of $\mathbf{B}$ is $COCl$, and the actual one should be oxalyl chloride $(COCl)_2$.

CHECKPOINT 1 PTS

The simplest formula of $\mathbf{B}$ is $COCl$, and the actual one should be oxalyl chloride $(COCl)_2$

$\mathbf{E}$ has a volatile odor, considering that $\mathbf{E}$ contains sulfur element; then $\mathbf{A}$ contains sulfur element. Considering that $\mathbf{A}$ is composed of four elements and is common, consider common short-period elements such as C, H, O, Cl.

CHECKPOINT 1 PTS

$\mathbf{E}$ has a volatile odor, considering that $\mathbf{E}$ contains sulfur element; then $\mathbf{A}$ contains sulfur element

Common four-element compounds containing sulfur are likely to be organic compounds; consider common ethanethiol CH_3CH_2SH and dimethyl sulfoxide $(CH_3)_2SO$. According to the mass fraction, $\mathbf{A}$ is $(CH_3)_2SO$, in which the mass fraction of sulfur is 41.04%.

CHECKPOINT 2 PTS

$\mathbf{A}$ is $(CH_3)_2SO$

Then, at this time, $\mathbf{E}$ is a volatile odorous liquid, then $\mathbf{E}$ is dimethyl sulfide $(CH_3)_2S$; considering the reaction temperature and reactants (ethanol, oxalyl chloride, dimethyl sulfoxide, producing dimethyl sulfide), it can be judged that the reaction is the Swern oxidation reaction. Therefore, the irritating odorous liquid $\mathbf{D}$ produced is acetaldehyde CH_3CHO.

CHECKPOINT 1 PTS

$\mathbf{E}$ is dimethyl sulfide $(CH_3)_2S$

CHECKPOINT 2 PTS

This reaction is the Swern oxidation reaction

CHECKPOINT 1 PTS

D is acetaldehyde CH_3CHO

In Swern oxidation, oxalyl chloride is converted into CO, CO_2, HCl, which requires the participation of a weak base to neutralize the generated HCl. The commonly used weak base is triethylamine $N(C_2H_5)_3$. According to the mass fraction, the weak base **C** is indeed triethylamine $N(C_2H_5)_3$.

CHECKPOINT 1 PTS

In Swern oxidation, oxalyl chloride is converted into CO, CO_2, HCl

CHECKPOINT 1 PTS

C is triethylamine $N(C_2H_5)_3$

Therefore, write the reaction equation:

$$C_2H_5OH + (CH_3)_2SO + (COCl)_2 + 2N(C_2H_5)_3 = 2N(C_2H_5)_3 \cdot HCl + CO_2 + CO + CH_3CHO + (CH_3)_2S$$

CHECKPOINT 2 PTS

$$C_2H_5OH + (CH_3)_2SO + (COCl)_2 + 2N(C_2H_5)_3 = 2N(C_2H_5)_3 \cdot HCl + CO_2 + CO + CH_3CHO + (CH_3)_2$$

The sum of the reaction coefficients is 11, only option D is correct.

At $300\ \mathrm{K}$, acetaldehyde is a gas and dimethyl sulfide is a liquid, so there are only three gases, option H is wrong.

CHECKPOINT 1 PTS

At $300\ \mathrm{K}$, acetaldehyde is a gas and dimethyl sulfide is a liquid

In Swern oxidation, $(COCF_3)_2$ or DCC can be used to replace oxalyl chloride, options I and J are wrong.

CHECKPOINT 1 PTS

In Swern oxidation, $(COCF_3)_2O$ or DCC can be used to replace oxalyl chloride